Assignment 05

Topic: Financial Market Data Investigation

Sample Data:

Source	Symbol	Time (IST)	LTP
Exchange	ABC (fictional)	10:15:02 AM	₹1,250.50
GFDL API	ABC (fictional)	10:15:01 AM	₹1,249.50
Client application	ABC (fictional)	10:15:00 AM	₹1,248.50

Investigation:

- Check symbol and instrument identifier: Confirm that ABC refers to same instrument in all three system.If one system maps the symbol to different instrument,display prices may differ

- Check timestrap and timezone : Make sure the timezone is known and compare observations from the same or closely aligned time

- Check market status: Confirm whether the market is open or closed and whether the displayed value is a current trade, previous close, or an older cached value

- Check real-time versus delayed data: Check whether the customer is supposed to receive live data or delayed data

- Check exchange and segment: Confirm that all observations refer to the same exchange and market segment, and to the same contract if the instrument is a derivative

- Check the type of price: Confirm that all systems show the same field, such as Last Traded Price (LTP), Close, Volume, Bid, or Ask.

Comparison Table:

What to compare	What to verify	Example evidence
Symbol / instrument	Same instrument in all systems	Symbol and instrument identifier
Exchange / segment	Same market and contract	Exchange and segment in request/configuration
Timestamp	Same time and timezone	Exchange time, API timestamp, client receive time
Market status	Correct session and current/stale value	Market session and last update time
Feed mode	Real-time or delayed setting	Client configuration or entitlement confirmation
Data type	Same field and meaning	LTP compared with LTP, not Close or Bid

Explain why a different value does not automatically prove the upstream data is incorrect:

Market prices can change over time. The systems may also refer to different instruments, exchange segments, timestamps, or data fields. The client may receive delayed data or display an older cached value. Therefore, the raw API response must be checked before deciding where the problem is

Collect evidence:

- The stock symbol and instrument identifier.
- The exchange and segment.
- The price shown in each system
- The time when each price was observed.
- The API response, with passwords and API keys removed.
- Whether the customer uses real-time or delayed data.
- Relevant error logs, if available.

LTP : Compare the last traded price and timestamp

OHLC: Check Open, High, Low, Close and the relevant trading period

Volume: Check whether volume is cumulative or for a specific interval

Bid/Ask: Compare bid and ask prices, quantities, and quote timestamps

Historical data: Check date range, interval, timezone, and field definition

Client Facing Explaination:

Hello Team,

Thank you for reporting the difference in market-data values. To investigate, please share the exact symbol and instrument identifier, exchange/segment, data field, and values with timestamps and timezone. Please also provide the sanitized raw API response, relevant logs or request ID, and confirmation of whether the configuration uses real-time or delayed data. Please remove API keys, tokens, passwords, and other confidential information.

We will compare the observations using the same instrument, field, and time, then check the configuration and client-side handling. If the mismatch remains reproducible, we will escalate it with the supporting evidence.

Regards,
Technical Support Team

Cocnclusion:

The correct approach is to verify the symbol, exchange/segment, timestamps, market status, feed mode, and data type. Compare the raw API response with the client display and collect evidence. Escalate only when the mismatch remains reproducible after these checks